

Mini Cases

Risk Management

2025/2026

Please upload your solutions and indicate which Mini Cases (MCs) you have solved before **May 27, 2026 at 7am**. To do so, go to the assignment section on Canvas and choose “Unit 1 Indication and Submission of MCs”. Then answer the true/false questions (Questions 1 to 6) and upload your solutions file (Question 7). **There will be no deadline extensions! Only complete submissions containing answers to the T/F questions and a solutions upload will be considered, your last complete attempt counts!** You can either write your solutions using software or scan your handwritten solutions. Please note that you may only submit one solutions file. If your solutions consists of more than one file (e.g. 1 pdf and 1 Excel file), please upload a .zip file. Uploaded files will be checked for plagiarism. The solutions in your file should be self-explanatory - not only for the instructor, but for other students as well. Indicate only Mini Cases you feel confident to present.

Important remark for the presentations: Make sure to provide a concise explanation of your solutions. In particular, when applying any formula from the lecture materials or other sources, you may be asked to explain the structure of your solutions and provide the intuition and underlying concepts behind the formula used.

Mini Case 3.1.

Consider a portfolio consisting of 15 assets with statistically independent default events (PDs, notionals, and simulated random variables can be found in "Mini_Case_3.1.xlsx" or "Mini_Case_3.1.Rdata").

- Both EAD and LGD are equal to 100%. Derive the expected loss rate and 1% VaR using Monte Carlo Simulation.
- Consider another scenario when EAD is equal to 100%, however LGD is 60%. Derive the expected loss rate and 1% VaR again using Monte Carlo Simulation and compare the results.

Mini Case 3.2.

Consider a portfolio consisting of $N=500$ assets with equal credit ratings. Assume that for every asset $i \in \{1, \dots, N\}$ the notional is 2000 USD, the EAD is equal to 100%, $PD_i=4.36\%$, $LGD_i=40\%$, $i \in \{1, \dots, N\}$ and LGDs are deterministic. Defaults are assumed to be statistically independent. Derive the expected value and standard deviation of the portfolio loss.

Hint:

The default indicator of asset $i \in \{1, \dots, N\}$ is $I_i \sim B(1, PD_i)$.

Extra exercise part for prospective QFin students and those who are interested:

(It is not for credit, the mini case can be indicated without solving this.)

Assume that the defaults are statistically dependent, with pairwise default correlations (i.e. correlation between pairs of default indicators I_i) of $\rho = Corr(I_i, I_j) = 0.008$. Derive the expected value and standard deviation of the portfolio loss.

Mini Case 3.3.

Consider a portfolio consisting of 10 assets where the asset values are jointly normally distributed with variance-covariance matrix Σ (Standard and Poor's credit ratings, notionals, and the Cholesky decomposition of the correlation matrix can be found in "Mini_Case_3.3.xlsx" or "Mini_Case_3.3.Rdata"). EAD is equal to 100% and the LGD is 70% for all assets. Use the average one-year default rates within the S&P rating classes (see in the slides) as the estimates of PDs. Simulate correlated random variables, then derive the expected loss rate and the 1% VaR using Monte Carlo Simulation (repeat 5,000 times).

Hint:

You can simulate standard normal random variables in Excel by drawing from a uniform distribution in $[0, 1]$ and applying the inverse standard normal CDF, i.e., $NORM.S.INV(RAND())$.

Mini Case 3.4.

Consider a portfolio consisting of 16 assets. The asset values are driven by one common factor plus individual, idiosyncratic factors. (PDs, notionals, and asset betas can be found in "Mini_Case_3.4.xlsx" or "Mini_Case_3.4.Rdata"). EAD is equal to 100% and the LGD is 70% for all assets. All assets have a volatility of 0.3 and the market volatility is 0.2. Calculate the factor loadings and then derive the expected loss rate, as well as the 1% VaR using Monte Carlo Simulation (repeat 2,500 times).

Mini Case 3.5.

Consider a portfolio of 500 correlated loans. The notional of each position is 2000 USD and is characterized by an EAD of 100%, LGD of 60% and PD of 4.36%. Assume that the diversity score for this class of obligors is 20%. Derive the expected loss rate and 1% VaR using Monte Carlo Simulation (repeat 3,000 times).

Hint:

Do not forget to account for the original portfolio.

Mini Case 3.6.

Consider a portfolio consisting of 10 assets with statistically independent default events (PDs and notionals can be found in "Mini_Case_3.6.xlsx" or "Mini_Case_3.6.Rdata"). All assets have the same EAD and LGD. EAD is equal to 100% but LGD is also a random variable that is uniformly distributed in $[0, 1]$ and is thus different in each run.

- First, derive the expected loss rate and 5% VaR using Monte Carlo Simulation ($K = 1000$) as in the Mini Cases before.
- Second, repeat the whole simulation 1,000 times such that you obtain a series of VaR estimates.

Hint:

In Excel you can repeat your analysis in the following way: First, you create a new sheet and in cell B1 reference to the VaR result you just calculated. Second, you select cells A1 to B1001 and use the "Data-Table Tool" in "What-If Analysis" (omit the row input cell and link the column input cell to an empty cell (e.g. C1).

- Third, analyze the histogram of the VaR estimates and calculate the standard deviation to obtain an estimate for the Monte Carlo error.

Hint:

If you are using Excel it might be necessary to submit this Mini Case in a separate Excel file.